Assignment 7

Q1.

```
#display login name and home directory

echo "LOGIN NAME: $USER"

echo "HOME DIRECTORY: $HOME"
```

```
2. Cat
3. Ls
4. Pwd
5. Exit
============
Select an option [1-5]: 5

Exiting. Thank you!

sunbeam@sunbeam-IdeaPad:~$ vim answ02.txt
sunbeam@sunbeam-IdeaPad:~$ vim vim.txt
sunbeam@sunbeam-IdeaPad:~$ bash vim.txt
LOGIN NAME: sunbeam
HOME DIRECTORY: /home/sunbeam
sunbeam@sunbeam-IdeaPad:~$
```

Q2.

```bash
#!/bin/bash

until [ "$choice" = "5" ]; do
    # Display the options
    echo "=== MENU ==="
    echo "1. Date"
    echo "2. Cal"
    echo "3. Ls"
    echo "4. Pwd"
    echo "5. Exit"
    echo "============"

    # Capture user input
    read -p "Select an option [1-5]: " choice
    echo ""

    # Switch-case structure evaluation
    case "$choice" in
        1)
            echo "Executing 'date':"
            date
            ;;
        2)
            echo "Executing 'cal':"
            cal
            ;;
        3)
            echo "Executing 'ls':"
            ls
            ;;
        4)
            echo "Executing 'pwd':"
            pwd
            ;;
        5)
            echo "Exiting. Thank you!"
            ;;
        *)
            echo "Error: Invalid selection. Try again."
            ;;
    esac
    echo ""
done
```

```
sunbeam@sunbeam-IdeaPad:~$ bash answ03.txt
answ03.txt: line 1: iread: command not found
Error: '' does not exist.
sunbeam@sunbeam-IdeaPad:~$ bash answ02.txt
=== MENU ===
1. Date
2. Cal
3. Ls
4. Pwd
5. Exit
============
Select an option [1-5]: 1

Executing 'date':
Wed Jun 24 07:17:43 PM IST 2026

=== MENU ===
1. Date
2. Cal
3. Ls
4. Pwd
5. Exit
============
```

Q.3

```
iread -p "Enter a files or directory name: " name

if [ ! -e "$name" ]; then
echo "Error: '$name' does not exist."
exit 1
fi

if [ -f "$name" ]; then
  echo "'$name'is a file."
  echo "size of the file :"
  ls  -lh "$name" | awk '{print $5}

  elif [ -d "$name ]; then
  echo "'$name is a directory."
  echo "content of the directory:"
  ls -p "$name"

   fi
```

Q4,

```
#!/bin/bash
# Find primes using pure bash arithmetic

START=2
END=100

echo "Prime numbers between $START and $END:"

for ((num=START; num<=END; num++)); do
    if [ $num -lt 2 ]; then continue; fi

    is_prime=1
    # Check divisibility up to the square root of the number
    for ((i=2; i*i<=num; i++)); do
        if [ $((num % i)) -eq 0 ]; then
            is_prime=0
            break
        fi
    done

    if [ $is_prime -eq 1 ]; then
        echo -n "$num "
    fi
done
echo ""

~
```

```
gratest: 32
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$ bash Assign04.txt
Prime numbers between 2 and 100:
2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83 89 97
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$
```

Q5

```
read -p "Enter first number:" num1
read -p "Enter second number:" num2
read -p "Enter third number:" num3

if (( num1 >= num2 && num1 >= num3))
then
echo "num1 Greatest : $num1"
elif ((num2 >= num1 && num2 >= num3))
then
echo "num2 greatest : $num2"
else
 echo "gratest: $num3"
 fi
```

```
these is a leap year: 2024
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$ vim Assign05.txt
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$ bash Assign05.txt
Enter first number:25
Enter second number:15
Enter third number:32
gratest: 32
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$
```

Q.6.

```
read -p "Enter a year" year

if ((year % 4 == 0))
then
echo "these is a leap year: $year"
else
echo "these is not aleap year :$year"
fi
~
~
```

```
number is negative: -58
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$ vim Assign06.txt
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$ bash Assign06.txt
Enter a year2024
these is a leap year: 2024
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$
```

Q7.

```
 read -p "Enter a number: " num
if((num < 0))
then
echo "number is negative: $num"
elif((num >= 0))
then
echo "number is positive: $num"
fi
~
~
~
~
```

```
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$ bash Assign07.txt
Enter a number: -58
number is negative: -58
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$
```

Q.8.

```bash

#!/bin/bash
echo -n "Enter number: "
read num

echo "Table of $num:"
for(( i=1 ;i<=10 ; i++))
do
echo $(expr $num \* $i )
done
```

```
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$ bash Assign08.txt
Enter number: 15
Table of 15:
15
30
45
60
75
90
105
120
135
150
```

Q.9.

```bash

i=1
until [ $i -gt $num ]
do
fact=`expr $fact \* $i`
i=`expr $i + 1`
done

echo "$num! = $fact"
```

```
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$ vim Assign9.txt
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$ vim Assign09.txt
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$ bash Assign09.txt
Enter number :20
20! = 2432902008176640000
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$
```

Q10.

```
Jun 24 18:57
sunbeam@sunbeam-IdeaPad: ~/Documents/osclasswork
#!/bash/bin

read -p "Enter number of term: " n
a=0
b=1
for ((i=0; i<n; i++))
do
echo -n "$a "
next=$((a + b))
a=$b
b=$next
done
echo ""
```

```
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$ vim Assign10
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$ vim Assign10.txt
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$ bash Assign10.txt
Enter number of term: 5
0 1 1 2 3
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$
```

Q11

```
echo "Enter Basic Salary:"
read basic

da=$(echo "$basic * 0.40" | bc -l)
hra=$(echo "$basic * 0.20" | bc -l)
gross=$(echo "$basic + $da + $hra" | bc -l)

echo "Gross Salary = $gross"
```

```
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$ vim Assign11
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$ bash Assign11
Enter Basic Salary:
2000
Gross Salary = 3200.00
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$ vim Assign11
```

Q.12.

```bash
#!/bin/bash

if [ $# -eq 0 ]
then
    echo "Usage: $0 filename"
    exit
fi

if [ -f "$1" ]
then
    echo "Last Modification Time:"
    ls -l "$1"
else
    echo "File does not exist."
fi
~
~
```

Q13.

```bash
#!/bin/bash

echo "Hidden files:"
ls -d .*
~
~
~
~
~
~
~
~
~
~
~
~
~
```

```
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$ vim Assign13
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$ bash Assign13
Hidden files:
.hidden.txt
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$ vim Assign13
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$
```

Q.14.

```bash
#!/bin/bash

echo "Executable files:"
find . -maxdepth 1 -type f -executable
```

Q.15

```bash
#!/bin/bash

echo "Enter first file name:"
read file1

echo "Enter second file name:"
read file2

if [ -f "$file1" ] && [ -f "$file2" ]
then
    tr 'A-Za-z' 'a-zA-Z' < "$file1" >> "$file2"
    echo "Contents appended successfully."
else
    echo "One or both files do not exist."
fi
~
~
~
~
~
~
~
~
~
~
~
```

```
Executable files:
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$ vim Assign15
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$ bash Assign15
Enter first file name:
vaibhavi.txt
Enter second file name:
Assign11.txt
One or both files do not exist.
```

Q.16

```
sunbeam@sunbeam-IdeaPad: ~/Documents/osclasswork
#!/bin/bash

echo "Welcome $(whoami)"
echo "Contents of Home Directory:"
ls ~
```

```
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$ vim Assign15
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$ vim Assign16
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$ bash Assign16
Welcome sunbeam
Contents of Home Directory:
 answ02.txt   Documents          'Git repo link for module.txt'        one       package.json        Public        script03.txt   todayexam     vd2      vim.txt
 answ03.txt   Downloads           google-chrome-stable_current_amd64.deb  osc-sdm   package-lock.json   script01.txt  snap           vaibhavi.txt  vd3      wpt
 Desktop      eclipse-workspace   Music                                  osfile    Pictures            script02.txt  Templates      vd1           Videos
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$
```

Q17.

```
sunbeam@sunbeam-IdeaPad: ~/Documents/osclasswork
#!/bin/bash

for ((i=1; i<=5; i++))
do
    for ((j=1; j<=i; j++))
    do
        echo -n "* "
    done
    echo
done

sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$ vim Assign16
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$ vim Assign17
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$ bash Assign17
*
* *
* * *
* * * *
* * * * *
sunbeam@sunbeam-IdeaPad:~/Documents/osclasswork$
```